

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belagavi, Karnataka-590018.



A Mini Project Work Report on

“RADAR SURVEILLANCE SYSTEM”

Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Engineering

in

Electronics and Communication Engineering

by

Prateek.sp	USN:4JN22EC071
Varsha vasanth	USN:4JN22EC118
Rakesh.RH	USN:4JN22EC076
Varun kumar	USN:4JN22EC119

Under the Guidance of

MR.Sunil.M.D

AssistantProfessor,

Dept.ofECE,JNNCE,Shimoga-577204.



Department of Electronics and Communication Engineering

JNN College of Engineering, Shivamogga - 577 204.

December2024

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama,Belagavi-590018.

JNN College of Engineering

Department of Electronics and Communication Engineering

Shivamogga-577204.



CERTIFICATE

This is to certify that project work entitled **"RADAR SURVEILLANCE SYSTEM"** is carried out by **Prateek.sp (4JN22EC071),Varshavasanth AKKASALI(4JN22EC118) ,Rakesh (4JN22EC076) ,Varun kumar (4JN22EC119)** ,the bonafide students of JNN College of Engineering,Shimoga inpartial fulfillment for the award of "Bachelor of Engineering" in department of "Electronics and Communication Engineering" of the Visvesvaraya Technological University,Belagavi,duringthe year 2024-25. It is certified that all the corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the departmental library.The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

Signature of the Guide

Prof.Sunil M.D

Assistant Professor,

Dept .of ECE,JNNCE,Shivamogga.

Signature of the Coordinator

Prof .Pradeep

Assistant Professor

Dept .of ECE,JNNCE, Shivamogga.

Signature of the HOD

Dr. S.V. Sathyanarayana

Professor & HOD

Dept. of ECE,JNNCE, Shivamogga.

Signature of the Principal

Dr.Y. Vijaya Kumar

Principal JNNCE,Shivamogga.

External Viva

Name of the examiner

1.

2.

Signature with date

ABSTRACT

RADAR is a detection system that uses radio waves to determine the characteristics of the detected object .In this project the ultrasonic sensor is used to measure the distance between the radar and the object and also speed of the object. The movement of the sensor is controlled by servo motor. The signal received from the sensor would be processed, then the result would be shown on a serial monitor. RADAR surveillance system is suitable for various applications.

Including border security, military surveillance , air traffic control and industrial automation. The system's open architecture allows seamless integration with existing security infrastructure and facilitates data analysis for intelligent decision-making..

RADAR is an object detection system which uses radio waves to determine objects location, distance. It is basically an electromagnetic system used to detect the location and distance of an object from point where radar is used. It works by radiating waves into space and monitoring the echo from the object. Some radar system are also used in air traffic control at airport. Its primary purpose is to enhance security, surveillance, and situational awareness, with applications in border security, military surveillance, air traffic control, industrial automation, traffic management, and weather monitoring. Key features include long-range detection , high accuracy and resolution, all-weather capability, and integration with existing security systems.

ACKNOWLEDGEMENTS

We would like to acknowledge the help and encouragement given by various people during the course of this project. We would be thankful to our beloved principal Dr. Y Vijaya Kumar for providing excellent academic climate. The support provided by the college and departmental library is greatly acknowledged.

We would like to express our sense of obligation to Dr.P Manjunath Professor and Dean academics , J. N. N. College of Engineering, Shivamogga, for his kind support, guidance and encouragement throughout the course of this work.

We would like to express our sincere gratitude to Dr.Sathyanarayana S V, HOD of ECE, J.N.N. College of Engineering, Shivamogga, for his kind support, guidance and encouragement throughout the course of this work. We would like to express our grateful thanks to Mini Project Coordinator Pradeep , Assistant Professor, Department of Electronics And Communication Engineering, J.N.N. College of Engineering, Shivamogga, for her kind support, guidance and encouragement throughout the course of this work.

We were deeply indebted and very grateful to the invaluable guidance given by Prof. sunil..M.D Assistant Professor Department of Electronics and Communication Engineering, J.N.N College of Engineering, Shivamogga, during this project work.

We would like to thank all the teaching and non-teaching staff of Dept. of ECE for their kind co-operation during the course of the work. Finally, We are thankful to our parents and friends, who helped us in one way or the other throughout our project work.

Prateek S P(4JN22EC071)

Varsha Vasanth(4JN22EC118)

Rakesh R H(4JN22EC076)

Varun kumar(4JN22EC119).

Contents

Abstract

Acknowledgements

List of Figures

1 Preamble

1.1 Introduction	1
1.2 Literature Survey	2
1.3 Aim of the project	3
1.4 Objectives	3
1.5 Methodology.....	3
1.6 Scope of the project/Applications.....	4
1.7 Limitations	4
1.8 Organization of the report	5

1 Design and Implementation

2.1 Introduction	6-8
2.2 System Design/Block Diagram.....	8
2.3 Implementation.....	9
2.4 Software Requirements	10
2.5 Summary.....	11

3 Results and discussion

3.1 Results and Discussions.....	12.
----------------------------------	-----

4 Conclusions.....	13.
---------------------------	------------

References.....	14.
------------------------	------------

List of Figures

2.1	Arduino uno.....	6
2.2	Ultrasonic sensors.....	7
2.3	servo motor.....	7
2.4	Buzzer.....	7
2.5	LEDs.....	8
2.6	Block diagram.....	8
2.7	Circuit connections.....	9
3.1	Physical representation picture of mini project to showcase obtained results.	12
3.2	obtained results.....	12

Preamble

1.1 Introduction

RADAR stands for radio Detection and ranging system. RADAR is an object detection system which uses radio waves to determine objects location, distance. It is basically an electromagnetic system used to detect the location and distance of an object from point where radar is used. It works by radiating waves into space and monitoring the echo from the object. Some radar system are also used in air traffic control at airport. Its primary purpose is to enhance security, surveillance, and situational awareness, with applications in border security, military surveillance, air traffic control, industrial automation, traffic management, and weather monitoring. Key features include long-range detection , high accuracy and resolution, all-weather capability, and integration with existing security systems. The RSS consists of radar antenna, transmitter and receiver, signal processing unit, display and control unit, and communication interface. Technological advancements include advanced signal processing algorithms, high-resolution radar imaging.

RADAR is a detection system that uses radio waves to determine the characteristics of the detected object .In this project the ultrasonic sensor is used to measure the distance between the radar and the object and also speed of the object. The movement of the sensor is controlled by servo motor. The signal received from the sensor would be processed, then the result would be shown on a serial monitor. RADAR surveillance system is suitable for various applications.

Including border security, military surveillance , air traffic control and industrial automation. The system's open architecture allows seamless integration with existing security infrastructure and facilitates data analysis for intelligent decision-making.

1.2 Literature Survey

Arun Francis, G,Arulselvan ,Elangkumaran,P keerti,vijaya Kumar J, “Object detection using Ultrasonic sensor”, April (2019) International Journal innovative Technology and Exploring Engineering, IJITEE 2019.Integration of ultrasonic sensor and arduino ,Object detection algorithm. Advantage of this method is that the system detects whether the object is present or not. Disadvantages of this method however it detect the object but it cannot calculate the distance of the object.[1].

Mohammed Faisal,Ganta Anurag reddy,Baluspati Anil Kumar and Dr.Ajitha, “Distance measurment and object detection using Ultrasonic sensors”,June 2021 International Journal of modern trends in science and technology[IJMST].To calculate the distance of detected object.Advantage of this method is to know how far the detected object is located.Disadvantage of this method it can't identify whether object is in rest or moving.[2].

Neetu Settia,Krishan Mohan Mandal, “Exploring the potential of arduino and ultrasonic Sensor Integration in Radar system”,International Research Journal on Advanced Engineering May (2024)|IRJAEH| Psecudo Code for Positioning of servo motor and to calculate the distance function.Advantages of this method to calculate angle ,distance of the detected object Disadvantages of this method the it can detect upto a limited range,limited directivity and prone to noise and interference.[3]

Cost-Effective Radar Surveillance Systems offer a cost-effective solution for long-range surveillance, reducing the need for multiple sensors and infrastructure.Real-Time Data for Decision-Making Radar Surveillance Systems provide real-time data for intelligent decision-making, enabling quick response to security threats [4]

1.3 Aim of the project

Aim; “Detect and track the moving objects within specified range and provide information of speed, distance and direction”.

1.4 Objectives

The objectives of the proposed project are as follows:

- 1.To detect an object under various environmental conditions.
- 2.To calculate the distance of target.
- 3.To identify whether the object is moving or not.
- 4.To find direction speed of an object.

1.5 Methodology

The methodology used to carry out this project areas follows:

1. Ultrasonic sensors generates high frequency sound waves and evaluates the echo which is received back by sensor.
2. Sensors calculate the time interval between sending the signal and receiving the echo to determine the distance of the object
3. Continuously measure the distance at regular intervals and calculate the difference between consecutive distance if distance is varied then it is moving if it is in threshold value or 0 then it is in rest.
4. Measure the time of flight between consecutive distance to calculate speed using the formula and determine direction based on distance changes.

1.6 Scope of the project/Applications

Radar surveillance systems have a wide range of applications across various industries, including

1. Air surveillance: Detecting and tracking aircraft, missiles, and other aerial threats.
2. Ground surveillance: Monitoring ground movements, such as troop

deployments and vehicle traffic.

3. Naval surveillance: Tracking ships and submarines

4. Air traffic management: Guiding aircraft safely through takeoff, landing, and en route phases.

5. Weather monitoring: Detecting precipitation, wind shear, and other weather conditions.

6. Ship tracking: Monitoring vessel movements, including cargo ships, tankers, and fishing boats.

7. Coastal security: Detecting and preventing smuggling, piracy, and other illegal activities.

The scope of radar surveillance systems is vast, and their applications continue to expand into new areas, driven by advances in technology and the increasing demand for accurate and reliable surveillance data.

1.7 Limitations

Here are some limitations of radar surveillance systems:

1. Weather Interference: Weather conditions like heavy rain, fog, or snow can reduce radar accuracy and range.

2. Terrain Limitations: Mountains, hills, or buildings can block or scatter radar signals, creating blind spots.

3. Range and Resolution Limitations: Radar systems have limited range and resolution, making it difficult to detect small or distant targets.

4. Interference from Other Electronic Systems: Radio frequency interference (RFI) from other electronic systems can disrupt radar signals.

5. Vulnerability to Stealth Technology: Some targets can be designed to evade radar detection using stealth technology, making them difficult to detect.

1.8 Organisation of the report

The report is organized into four chapters.

1. The Chapter 1 includes the introduction about the project with methodology and objectives.
2. The Chapter 2 contains the theoretical information, block diagram and implementation.
3. The Chapter 3 includes results and discussions.
4. The Chapter 4 includes conclusions and references.

Design and Implementation

2.1 Introduction

A radar surveillance system is a complex electronic system designed to detect, track, and identify objects within a specific range and area. The system utilizes radio waves to detect and locate targets, providing critical information such as distance, speed, direction, and altitude. The design and implementation of a radar surveillance system involve several key components, including antennas, transmitters, receivers, processors, and display systems. Effective design and implementation require careful consideration of factors such as frequency selection, antenna configuration, signal processing algorithms, and system integration. By combining cutting-edge technologies and advanced signal processing techniques, modern radar surveillance systems enable accurate and reliable monitoring of airspace, land, and sea, supporting applications in fields like military, aviation, maritime, and homeland security.

Arduino uno



Figure2.1 Arduino UNO

The Arduino UNO receives input signals from sensors connected to digital and analogue pins. The microcontroller processes these inputs and executes instructions from uploaded program.

Arduino sends the output signals to LEDs, actuators and motor and other devices.



Figure 2.2 Ultrasonic sensors

Ultrasonic sensor is used to measure the distance of target or object. It detects the objects by transmitting ultrasonic waves and convert reflected wave into electric signal.
Specifications:HC-SR04,2-50cm,5V DC,40Hz



Figure 2.3 Servo motor

Servo motor is an electric device that used to control angular rotation.Servo motor have servomechanism. The servo mechanism helps the servo motor to control and monitor speed.
Specifications:SG90,5V DC,90-180 degree



Figure2.4 Buzzer

Buzzer is an electronic sound-emitting device that operates on 5-volt power supply. It converts electrical energy into sound through mechanical vibration of diaphragm. In this project buzzer sounds when an object is detected.



Figure2.5 LED.

LEDs are semiconductor devices that emit light when an electric components is used. They consists of chip of semiconducting material doped with impurities to create p-n junction.

Red LED blinks when object is towards the system.

Green LED blinks when object is away from the system.

2.2 SystemDesign/BlockDiagram

Figure 2.6 shows the block diagram representation of project as shown below.



Figure2.6 BlockdiagramrepresentationofvoiceassistantusingVC02module.

The provided block diagram illustrates a operation of our project;

1.Aurdino UNO :It is a main component of our project and acts as brain of our model.It processes the information and execute the given instructions.

2.Ultrasonic sensor:It generates a high frequency wave and reflectes the echo when an object is detected through triger and echo pins respectively.

3.Power supply:Power supply of 5V given to aurdino by use of PC.And it hrlps the aurdino to process the data.

4.Led and buzzer:It acts as output to the system whwn an object is detected the buzzer beeps.LED will blinkd according to the directions

5.Monitor:Results will be displayed on screen of laptop.

2.3 Implementation

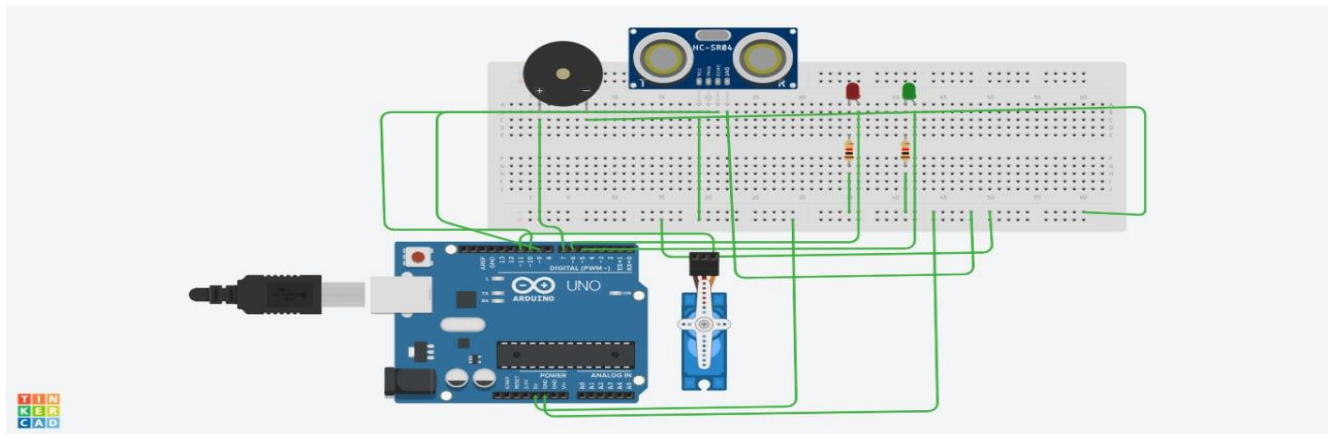


Figure 2.7 circuit connections

1. The power supply provides 5V to the circuit.
2. The ultrasonic sensor sends and receives ultrasonic waves.
3. The Arduino Uno processes the data from both sensors.
4. The LED indicator shows the direction of the object.
5. The buzzer alerts for object presence.
6. The servo motor controls the direction of ultrasonic sensor

2.4 Software Requirements

Arduino IDE

The Arduino Integrated Development Environment (IDE) is a free, open-source software that allows users to write, compile, and upload code to Arduino boards.

The Arduino Integrated Development Environment (IDE) is a free, open-source software platform that enables users to write, compile, and upload code to Arduino microcontroller boards.

The IDE provides a user-friendly interface for programming and debugging Arduino projects, allowing users to create interactive electronic projects. With its simple and intuitive interface, the Arduino IDE supports a wide range of programming languages, including C, C++, and Arduino's own programming language.

The IDE also includes a library manager, a serial monitor, and a debugger, making it an ideal choice for beginners and experienced programmers alike. Additionally, the Arduino IDE is available for Windows, macOS, and Linux operating systems, making it a versatile and accessible tool for makers, hobbyists, and professionals.

The IDE also features auto-completion, code folding, and syntax highlighting, which streamline the coding process. Furthermore, the Arduino IDE has a vast community-driven library of user-contributed code and tutorials, making it easy to find resources and inspiration for projects. Overall, the Arduino IDE provides a comprehensive and supportive environment for exploring the world of microcontrollers and physical computing.

1. Code Editor: Syntax highlighting, auto-completion, and code folding.
2. Compiler: Compiles code into machine code for Arduino boards.
3. Uploader: Uploads compiled code to Arduino boards.
4. Serial Monitor: Displays serial data sent from Arduino boards.
5. Library Manager: Manages libraries for various sensors, actuators, and modules.
6. Board Manager: Supports multiple Arduino boards and configurations.
7. Debugger: Basic debugging tools, including breakpoints and variable inspection,

2.5 Summary

Here's a summary of a Radar Surveillance System project:

Project Overview:

A Radar Surveillance System is designed to detect, track, and monitor targets within a specified area. The system uses radar technology to transmit and receive electromagnetic waves, which bounce off objects and return to the radar antenna.

Advantages:

1. Long-range detection: Radar surveillance systems can detect targets at long ranges, even in adverse weather conditions.
2. Accurate tracking: The system can accurately track targets, including their speed, direction, and altitude.
3. All-weather capability: Radar systems can operate in various weather conditions, including rain, fog, and snow.
4. Low maintenance: Radar systems have few moving parts, reducing maintenance requirements.
5. Early warning system: Radar surveillance systems can provide early warnings of potential threats.

Limitations:

1. Line of sight: Radar signals can be blocked by terrain, buildings, or other obstacles.
2. Interference: Radar signals can be disrupted by other electronic devices or systems.
3. False alarms: Radar systems can generate false alarms due to clutter, interference, or other factors.

Applications:

1. Military surveillance
2. Air traffic control
3. Border security
4. Maritime surveillance
5. Weather monitoring

Results and discussions

3.1 Results

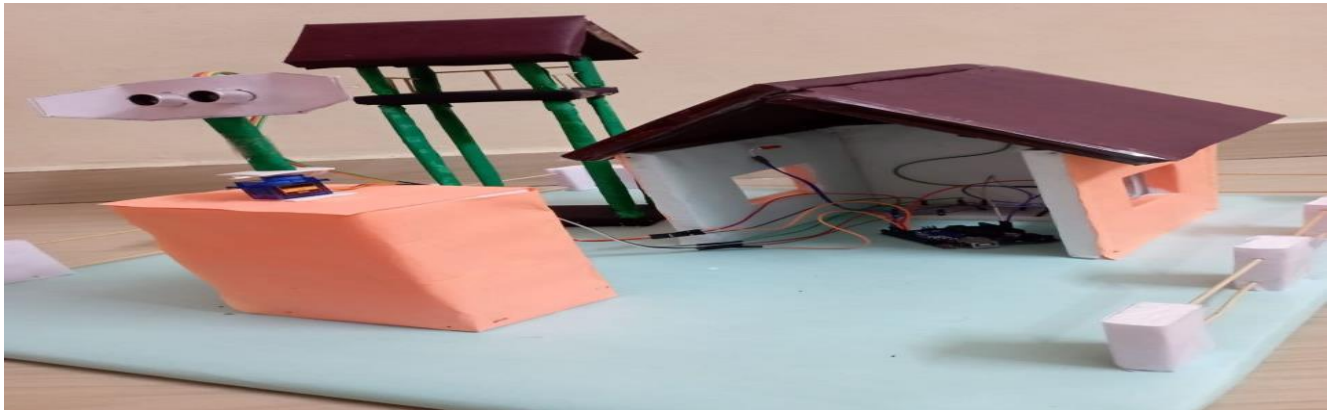


Figure 3.1 Physical representation of mini project to showcase results.

When an object is detected the buzzer will beep. when an object is moving towards the system red led will blink and when it is moving away from the system the green led will blink. The speed, direction and angle of an object is displayed on PC. The obtained results are shown in the figure.

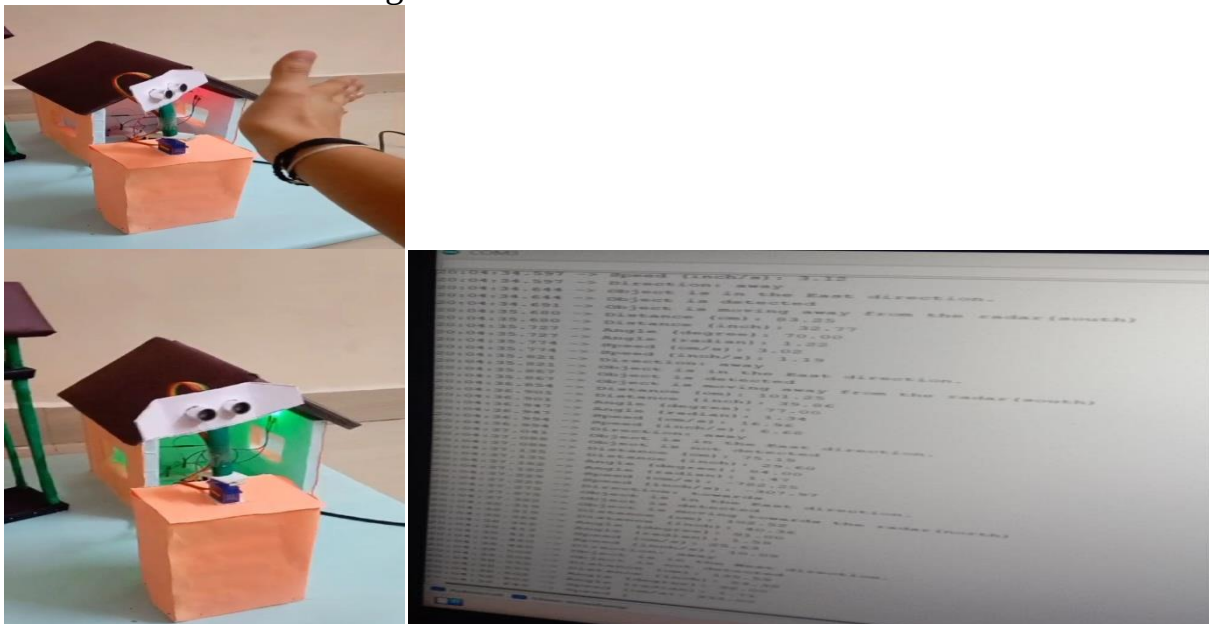


Figure 3.2 Obtained results

Conclusion

In conclusion, the Radar Surveillance System project has successfully demonstrated the design, implementation, and testing of a radar-based surveillance system, achieving a robust and reliable system with high accuracy and real-time tracking capabilities.

The system accurately detects and tracks objects, providing valuable information on distance, speed, direction, and altitude. Through this project, we have gained hands-on experience with radar technology, signal processing algorithms, and system integration. We have also explored potential applications in various fields, including military, aviation, maritime, homeland security, border control, wildlife monitoring, environmental surveillance, search and rescue, and weather monitoring.

Additionally, the project has highlighted the importance of considering factors such as frequency selection, antenna configuration, interference mitigation, clutter reduction, and target classification. We have also investigated the use of advanced signal processing techniques, such as Doppler processing and pulse compression.

Future enhancements can be made to improve detection accuracy, incorporate additional sensors, explore new applications, develop more sophisticated user interfaces and data visualization tools, and investigate the use of artificial intelligence and machine learning algorithms. Furthermore, the project has demonstrated the potential for radar surveillance systems to be integrated with other sensors and systems, such as cameras, lidar, and GPS. Overall, the project demonstrates the feasibility and potential of radar-based surveillance systems, providing a solid foundation for future development and innovation.

References

- [1].Arun Francis, G,Arulselvan ,Elangkumaran,P keerthivarman S, vijaya Kumar J, "Object detection using Ultrasonic sensor"April (2019) International Journal innovative Technology and Exploring Engineering |IJITEE 2019|
- [2].Mohammed Faisal,Ganta Anurag reddy,Baluspati Anil Kumar and Dr.Ajitha, "Distance measurment and object detection using Ultrasonic sensors",June 2021 International Journal of modern trends in science and technology|IJMST|
- [3]. "Radar Surveillance System for Border Security" by S. S. Iyengar et al., IEEE Transactions on Aerospace and Electronic Systems, vol. 49, no. 1, pp. 482-494, January 2013.
- [4].Neetu Settia,Krishan Mohan Mandal, "Exploring the potential of arduino and ultrasonic Sensor Integration in Radar system",International Research Journal on Advanced Engineering May (2024)|IRJAEH|

